



Lesson 3: Regression

Technologies for Artificial Intelligence

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Before we start

- Exercise 2, which we discuss now
- The pipeline you built is the one we fit models into from now on

Today: the first real model

- The model and what it claims
- The exact solution, and why it sometimes fails
- Gradient descent
- Curves, and the price of flexibility
- Ridge and Lasso

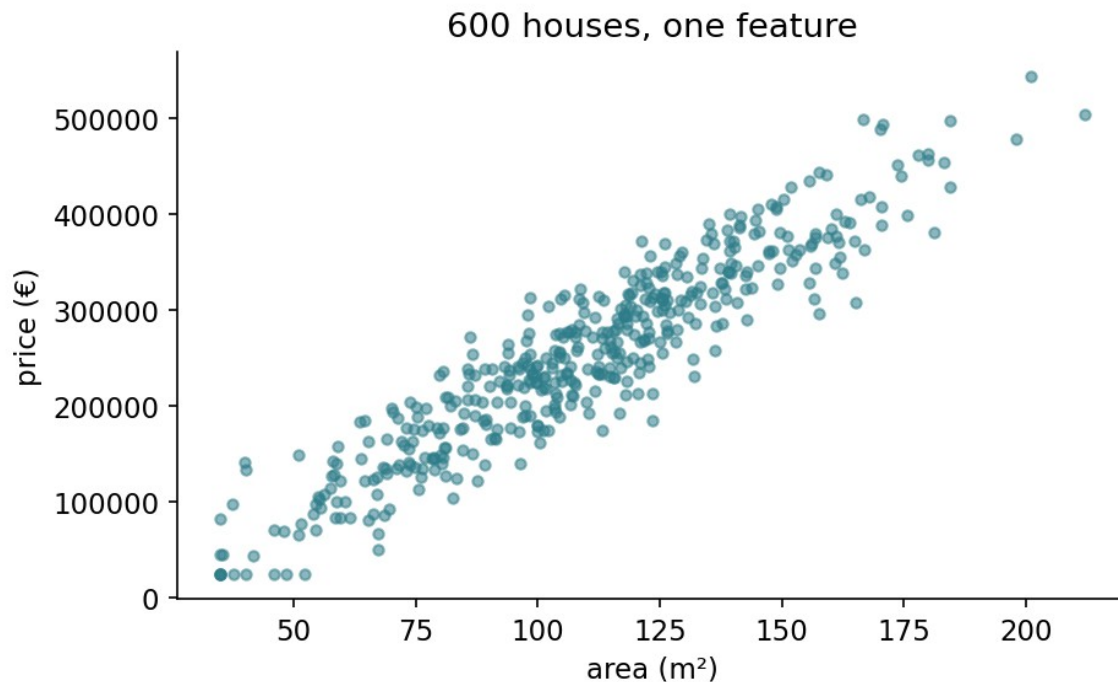
One dataset, all lesson

600 houses. Six measurements. A price.

And we know the coefficients that generated it.

In code: `train_test_split(X, y, test_size=0.25)` keeps 150 houses for testing.

What we are trying to predict



What a linear model claims

Each feature contributes a fixed amount per unit, and the contributions add up: $\hat{y} = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$, one coefficient w per feature plus an intercept b .

- Every square metre: the same 2,400 €
- Every kilometre out: the same –6,500 €

Fitting means minimising something

We need one number saying how badly a candidate model is doing.

Adding up the errors fails: $+50,000$ and $-50,000$ cancel to zero.

Why squared, not absolute

- **Differentiable everywhere:** no corner at zero
- **Punishes one large error more** than several small ones
- Under Gaussian noise it is **maximum likelihood:** the coefficients that make the observed prices most probable

The cost function: mean squared error

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2$$

Trying $\hat{y} = 2,400 \cdot \text{area} + 45,000$

Area (m ²)	True (€)	Predicted (€)	Error $\hat{y} - y$	Error ²
80	240,000	237,000	-3,000	9.0×10^6
120	320,000	333,000	+13,000	1.69×10^8
200	540,000	525,000	-15,000	2.25×10^8

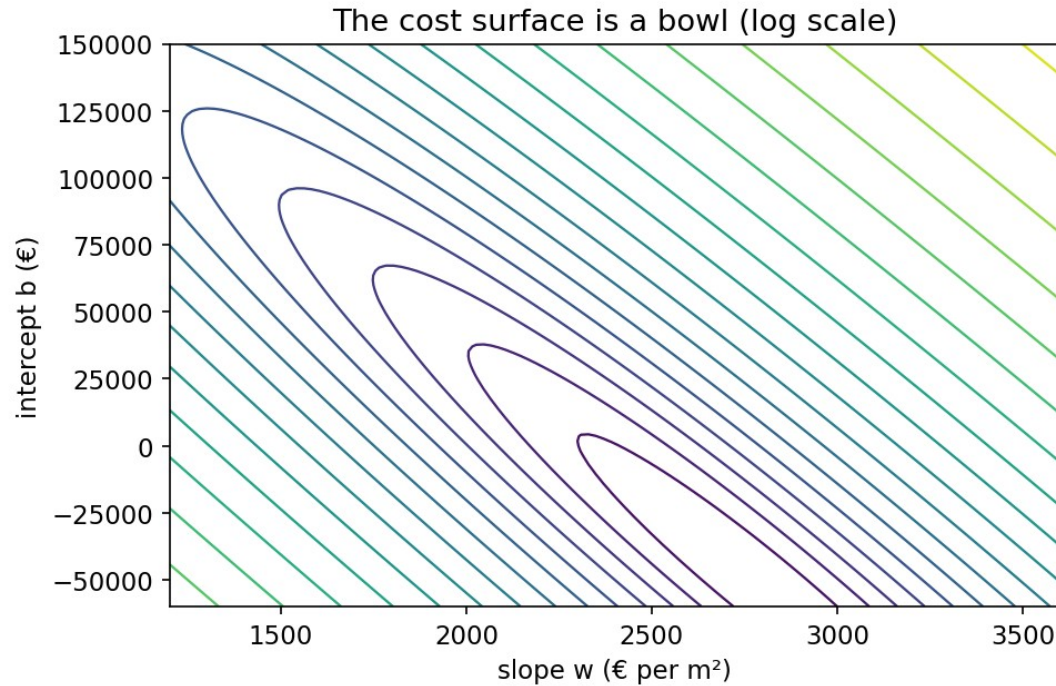
Squared error spends its attention on the worst house

- The 3,000 € error contributes about **2%** of the total
- The two large errors contribute the other **98%**
- Halving the 15,000 € error cuts the cost by **42%**
- In code: `mean_squared_error(y, y_pred)`, the mean of the squared errors without the $\frac{1}{2}$

A cost needs something to be compared against

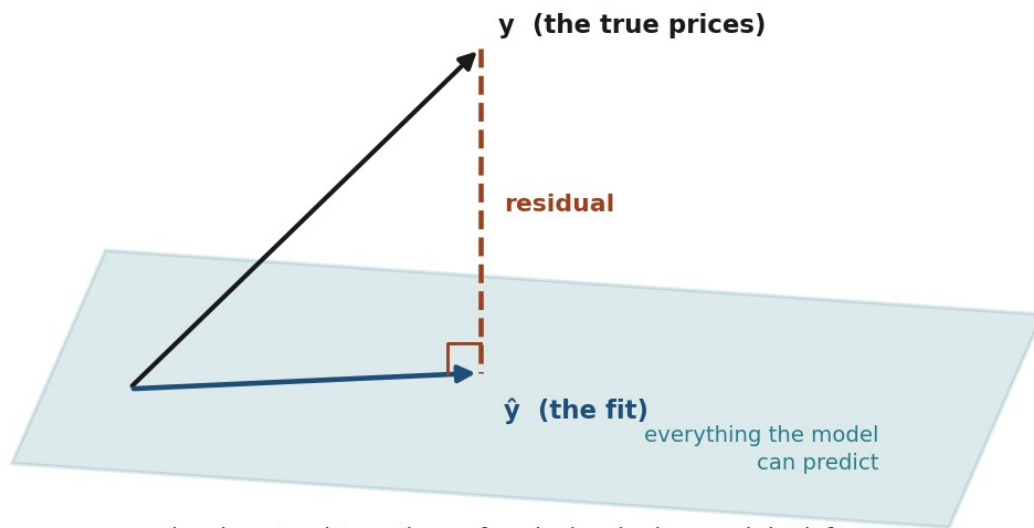
- Trivial answer: **always predict the mean**,
`DummyRegressor()`
- Root mean squared error, $RMSE = \sqrt{MSE}$, on the 150 test houses: **96,440 €** for the mean, **20,341 €** for the model
- $R^2 = 1 - \Sigma(y - \hat{y})^2 \div \Sigma(y - \bar{y})^2 = \mathbf{0.956}$
(`r2_score`): 0 is the mean, 1 is perfect
- Quote the euros too: R^2 depends on *this* test set's variance

The cost is a bowl



The picture behind the exact solution

Least squares is a projection



the closest point on the surface is the shadow, and the leftover error is perpendicular, or you could have done better

The normal equation

$$\theta = (X^T X)^{-1} X^T y$$

The same equation, by hand

Quantity	Value
the three houses	80, 120, 200 m ² → 240k, 320k, 540k
X, and θ	a column of 1s (for b) beside the areas; $\theta = (b, w)$
$X^T X$	rows (3, 400) and (400, 60,800)
$X^T y$	(1,100, 165,600)
determinant	22,400
slope w	2,536 €/m² (truth 2,400)
intercept b	28,600 €

And it really is the minimum

- The second derivative is $X^T X$, never negative in any direction
- So the cost is **convex**: one bottom, no local traps
- Any stationary point (zero gradient) is *the* answer, not *an* answer
- In code: `LinearRegression().fit(X, y)` solves the same problem

So why learn anything else?

- $(X^T X)^{-1}$ costs about n^3 operations
- Two columns carrying one fact → **no inverse exists**
- Nearly-redundant columns → an inverse that is useless

How stretched is the valley?

The **condition number** κ = steepest curvature $\div$ shallowest curvature: how many times steeper the steepest direction of the bowl is than the shallowest.

- Housing features as recorded: **285**
- The same features, standardised: **3.4**
- Add `area_sqft` beside `area_sqm`: **2,286**

The picture behind gradient descent

You are on a hillside in fog.

You cannot see the valley, but you can feel which way the ground slopes.

So: step against the slope, by an amount set by the **learning rate** α , and repeat until the cost stops falling.

The update rule

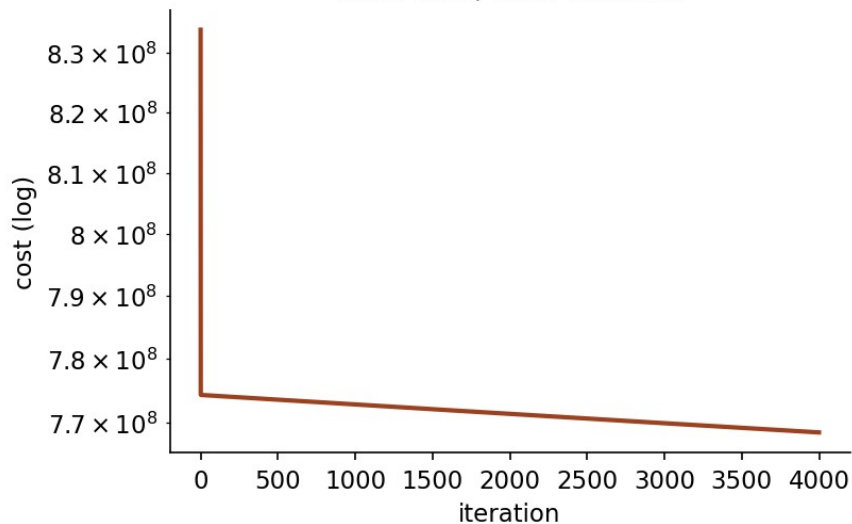
$$w \leftarrow w - \alpha \frac{\partial J}{\partial w}, \quad \frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_i (\hat{y}^{(i)} - y^{(i)}) x_j^{(i)}$$

One step on two houses, from $w = 0$ and $b = 0$

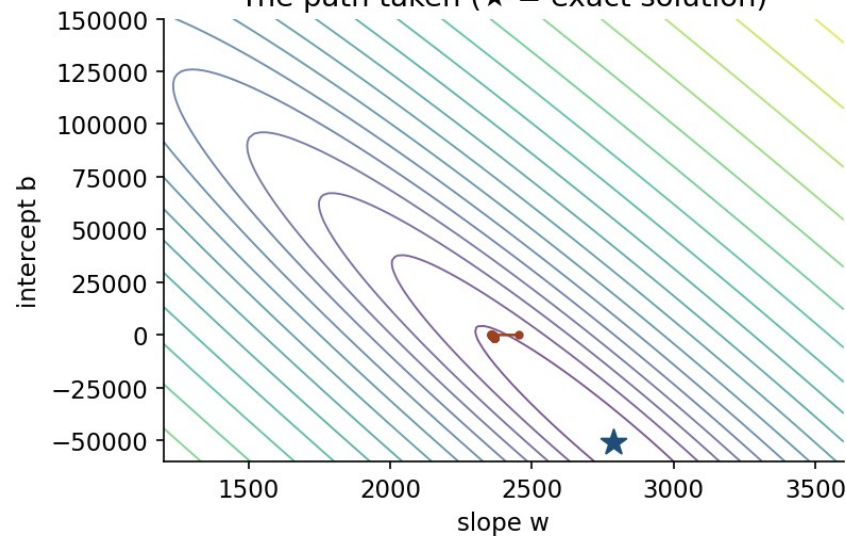
80 m ² at 240k, 200 m ² at 540k; $\alpha = 10^{-5}$	Gradient	After one step
slope w	-63,600	0.636
intercept b	-390	0.0039

The path it takes

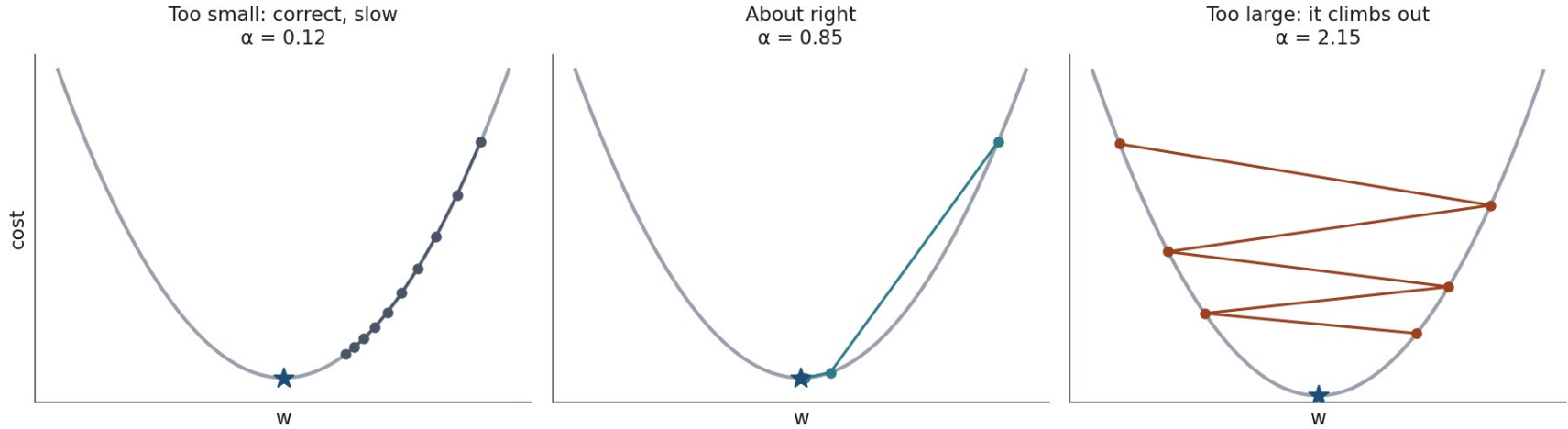
Cost falls, then flattens



The path taken (★ = exact solution)



Three learning rates



Notebook 1, live

section, and its slides	you call (→ the number to come back with)
one feature, 4-5	<code>train_test_split</code> → 450 houses to fit, 150 to test
the exact answer, 14-17	the normal equation in NumPy, then <code>LinearRegression</code> → 2,785 €/m ² , agreeing to 1.82e-12
the iterative answer, 20-24	NumPy only: gradient descent → the same answer, slowly
all six features, 49-50	<code>LinearRegression</code> → area 2,421, bedrooms -16.6%
the baseline, 12	<code>DummyRegressor(strategy="mean")</code> , <code>r2_score</code> → 96,440 € against 20,341 €, R ² 0.956

Break

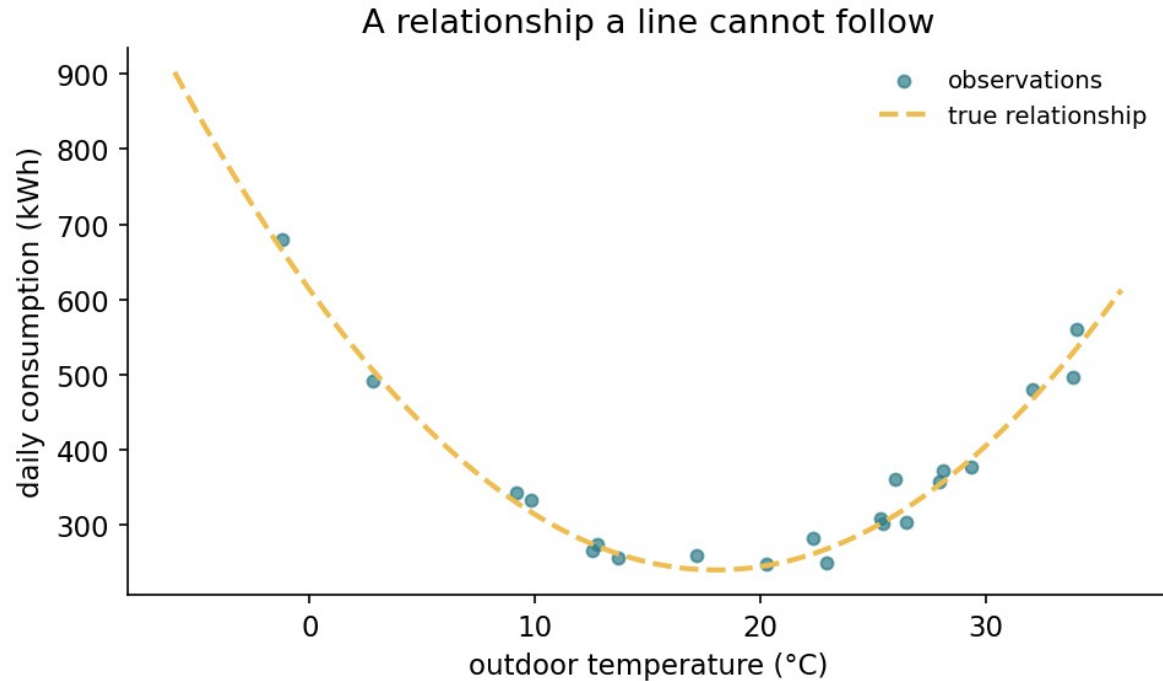
What if the relationship bends?

Energy consumption against temperature: heating in the cold, cooling in the heat.

No straight line follows that.

In code: `PolynomialFeatures(degree=2)` adds the squared temperature, t^2 , as a second column.

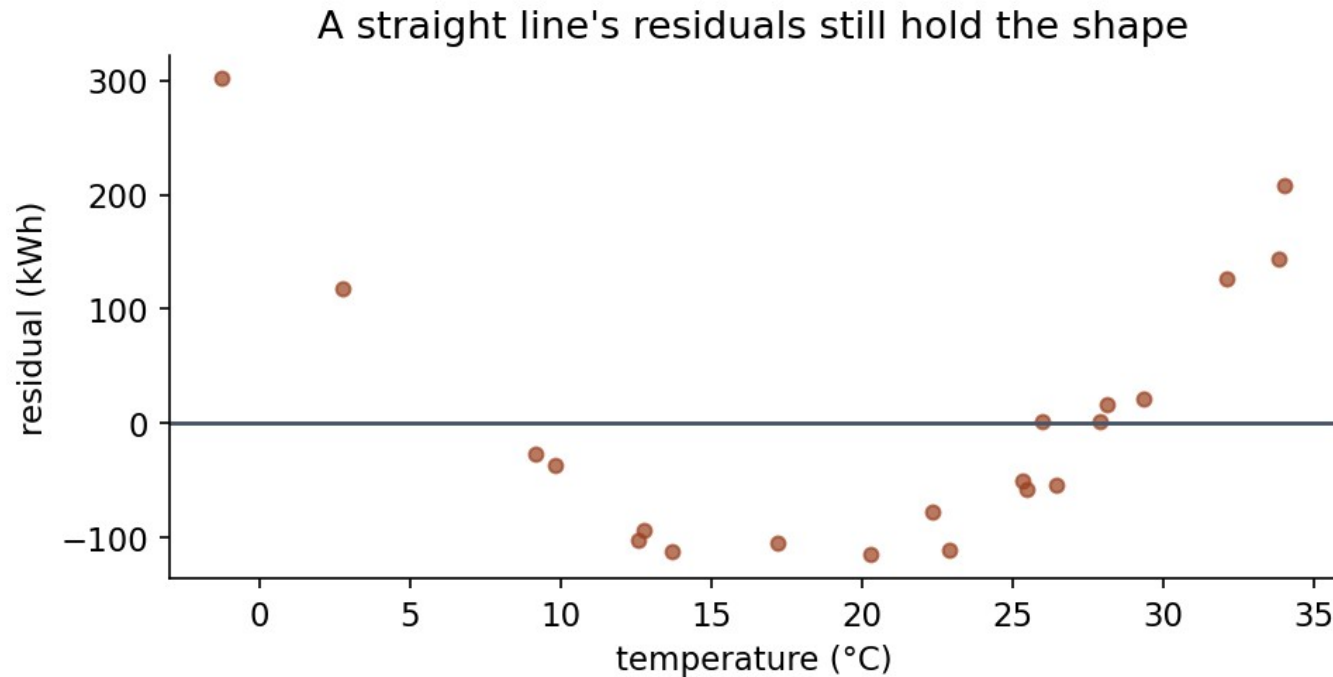
A relationship a line cannot follow



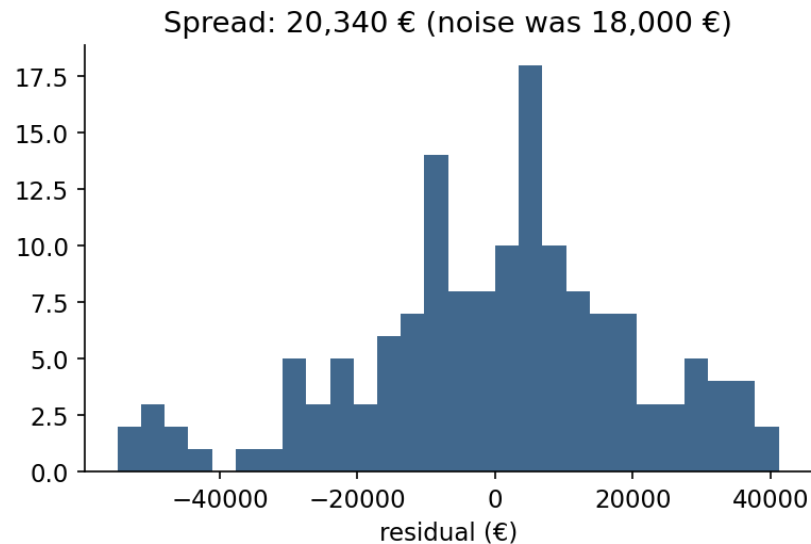
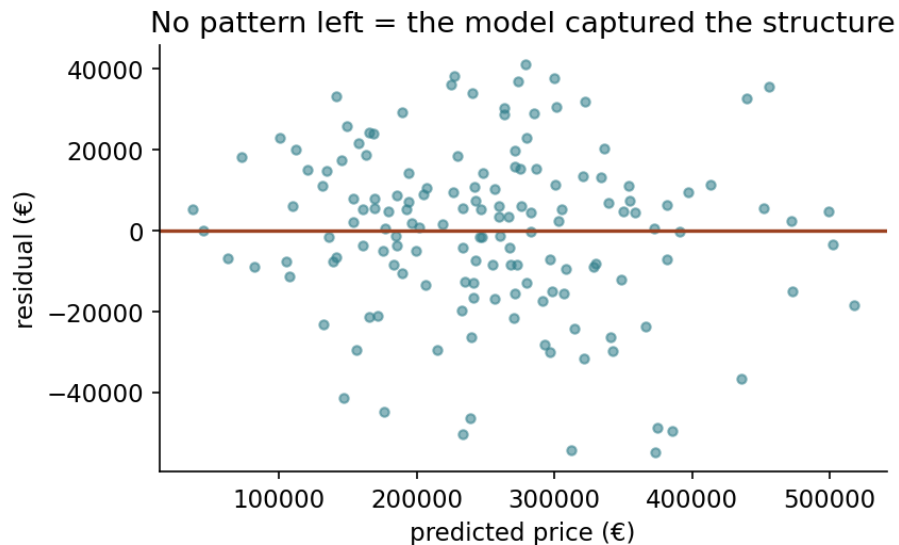
“Linear” means linear in the coefficients

$$\hat{y} = w_1 t + w_2 t^2 + b$$

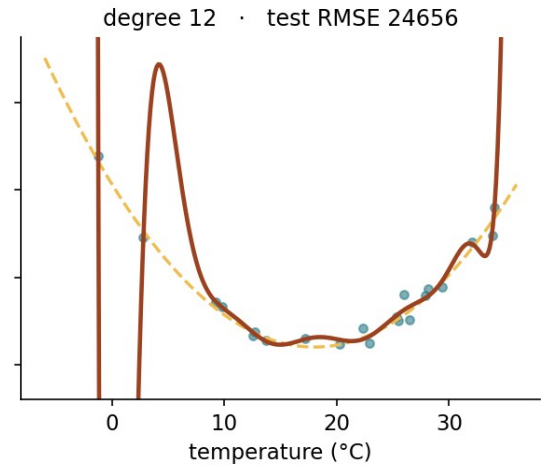
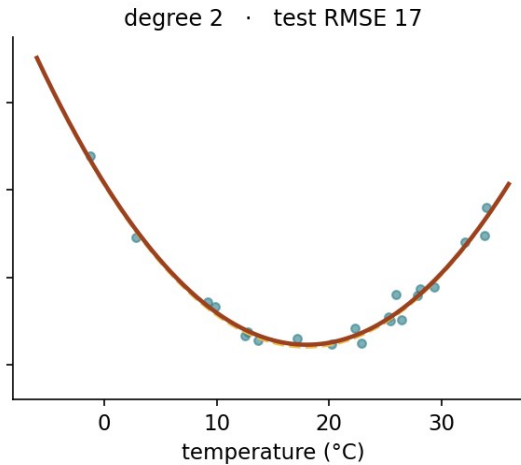
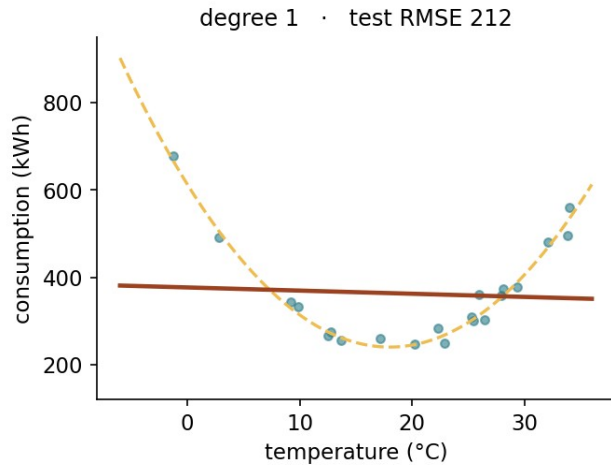
The straight line's residuals keep the shape



What “nothing left” looks like



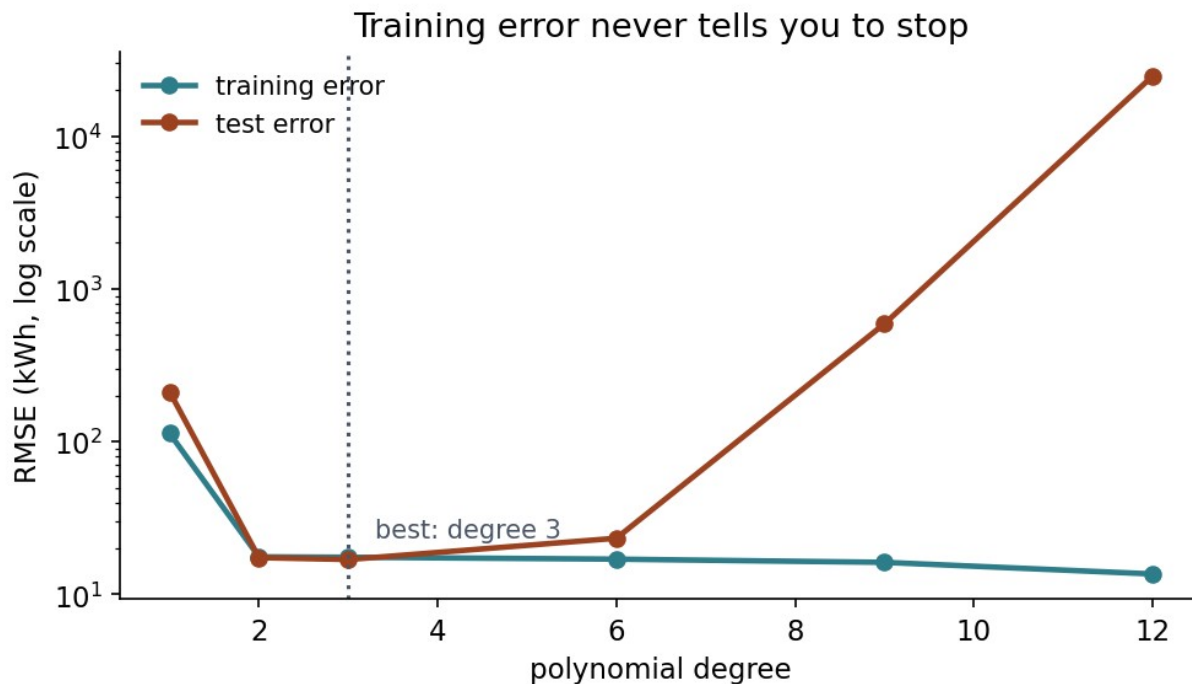
So use more flexibility?



Root mean squared error (RMSE), by degree

Degree	Train RMSE (kWh)	Test RMSE (kWh)
1	113.3	212.1
3	17.5	16.9
9	16.2	590.1
12	13.6	24,655.7

The gap is the overfitting



Notebook 2, live

section, and its slides	you call (→ the number to come back with)
a relationship that bends, 27-28	<code>train_test_split</code> → 21 days to fit, 9 to test
linear in the coefficients, 29-31	<code>PolynomialFeatures</code> , <code>StandardScaler</code> and <code>LinearRegression</code> , chained by <code>make_pipeline</code>
too much flexibility, 32-34	<code>mean_squared_error</code> , square-rooted → test RMSE 16.9 at degree 3, 24,655.7 at 12
why the wiggles appear, 36	the fitted coefficients → largest 3,097,038,010 at degree 12

Why flexibility goes wrong

Largest coefficient, degree 2: **411**

Largest coefficient, degree 12: **3,097,038,010**

Charge for size

$$J_{\text{ridge}} = \text{MSE} + \lambda \sum_j w_j^2 \quad J_{\text{lasso}} = \text{MSE} + \lambda \sum_j |w_j|$$

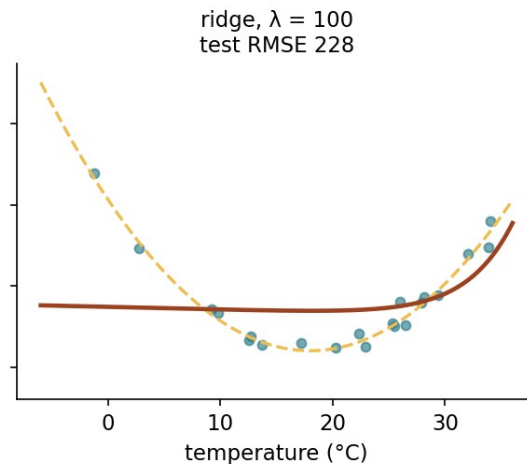
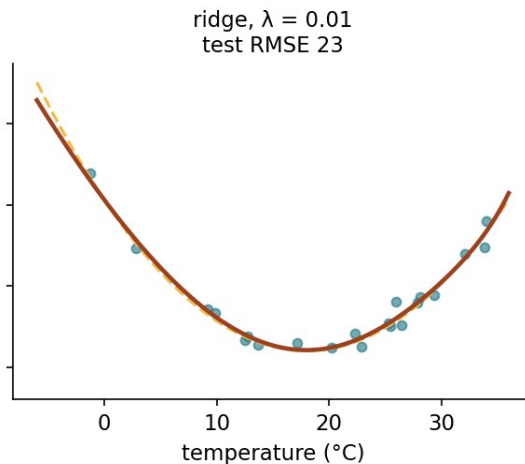
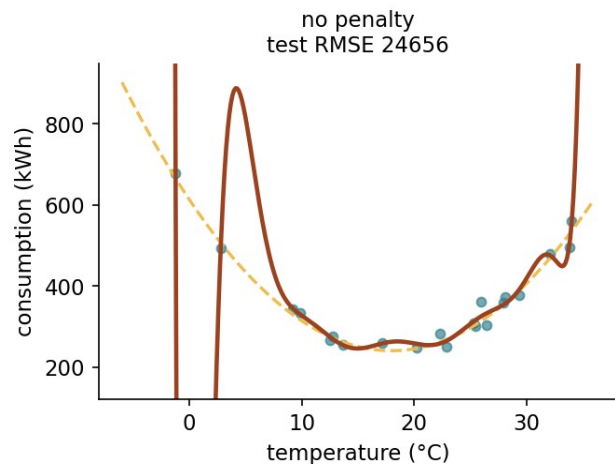
Two rules that are not optional

- The **intercept is never penalised**: it is not a claim about any feature
- Features **must be scaled first**, or the penalty is arbitrary
- In code: `make_pipeline(StandardScaler(), Ridge(alpha=0.01))`; scikit-learn calls λ alpha

What a penalty buys

Model	Train RMSE	Test RMSE	max w
none	13.6	24,655.7	3,097,038,010
$\lambda = 0.01$	18.0	22.8	365
$\lambda = 1$	44.5	105.2	156
$\lambda = 100$	96.0	227.6	6

The unpenalised fit leaves the picture



Too flexible, too rigid



Ridge has an exact solution too

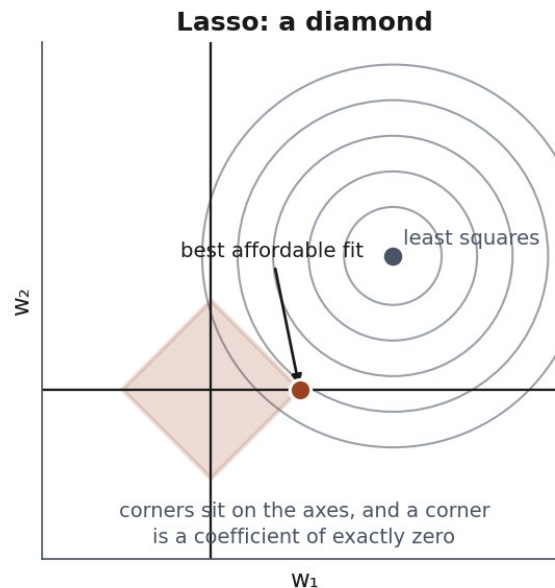
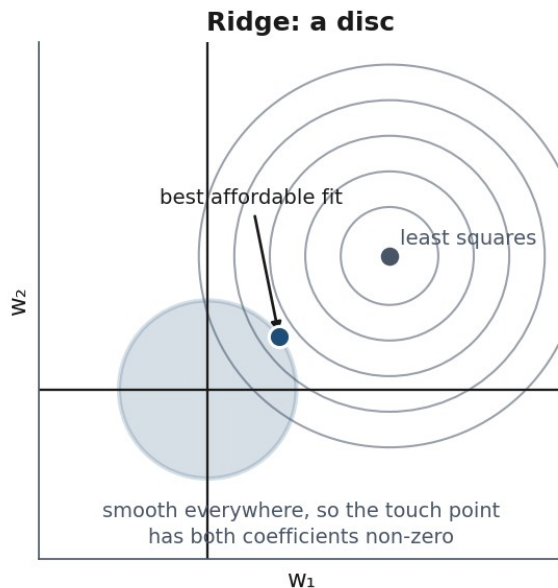
$$\theta = (X^{\top} X + \lambda I)^{-1} X^{\top} y$$

Why that one change fixes it

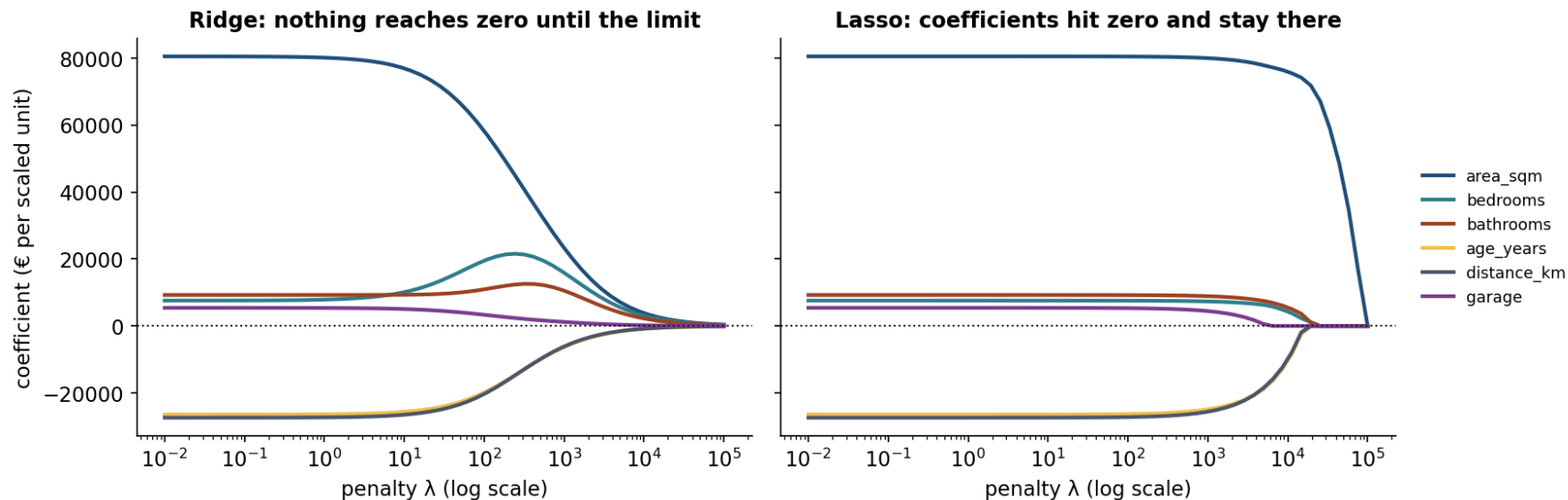
- Least squares fails when a direction costs nothing: the valley is flat
- The penalty makes **every** direction cost something
- Every eigenvalue of $X^T X$ shifts up by λ , so none of them can be zero

The picture behind Ridge vs Lasso

The budget you can afford has a shape



Ridge shrinks, Lasso selects



The order Lasso drops features

λ	Features kept
1,000	all six
10,000	five: garage dropped
20,000	three: area_sqm, bedrooms, bathrooms
40,000	one: area_sqm

The case Ridge was invented for

area_sqm and area_sqft: the same measurement,
two units.

Correlation: **0.999999**

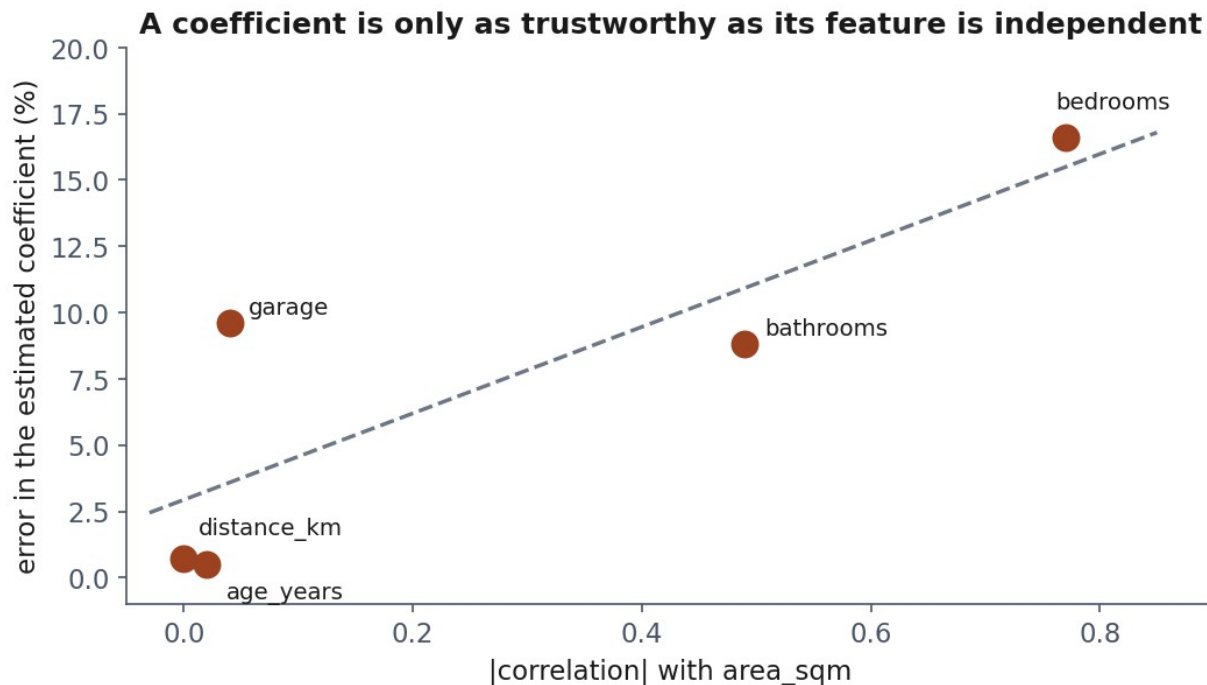
The coefficients are noise

Split	area_sqm	area_sqft
0	9,261	-640
1	14,535	-1,127
2	45,307	-3,989
3	39,061	-3,402

What a coefficient actually says

The change in the prediction for a one-unit change in that feature, **with every other feature held fixed**.

When can you trust a coefficient?



Notebook 3, live

section, and its slides	you call (→ the number to come back with)
Ridge on the degree-12 fit, 37-41	<code>Ridge(alpha=0.01)</code> after <code>PolynomialFeatures</code> and <code>StandardScaler</code> → largest coefficient 365
regularisation paths, 44-46	<code>make_pipeline(StandardScaler(), Ridge(alpha=...))</code> , the same with <code>Lasso</code> → only area left at 40,000
the collinear case, 47-48	<code>LinearRegression</code> against <code>Ridge(alpha=10)</code> on four splits → 9,261 to 45,307, then about 40,000

What we did today

- A model that claims a fixed amount per unit
- An exact solution (a projection), and when it fails
- An iterative one: walk downhill, mind your stride
- Flexibility helps until it does not
- Charging for size buys generalisation, and readable coefficients
- **365** against billions: what a penalty of 0.01 costs, and buys

Homework: we discuss it on Friday 23 October

Exercises/03_regression.md

Fit, regularise, and **explain which coefficients
you believe.**